

DataViz. (BSCS4001) May 2026 Project

Group-3

A Data-Driven Study of E-Commerce Platform in Brazil: Orders, Delivery & Customer Satisfaction

Team Details

Name	Student Email ID
Somsubhra De	22f3002680@ds.study.iitm.ac.in
Durga Prasad S	22f3001307@ds.study.iitm.ac.in
Balasurya K	22f3002744@ds.study.iitm.ac.in
Matlin Jeleshiya D	22f2000506@ds.study.iitm.ac.in
Sourish Sur	21f1001900@ds.study.iitm.ac.in

submitted to



BS in Data Science & Applications

Indian Institute of Technology Madras

Chennai 600036, INDIA

Course Faculty: Prof. Venkatesh Rajamanickam

Instructor: Rohit Kumar

Declaration

We hereby declare that the work presented in this report is our own genuine work, carried out as part of project in the Data visualization coursework. The work is original and has not been copied from any other source. All team members unanimously agree that each has contributed equally to the completion of this project. To ensure transparency, accountability and consistent progress throughout the project, we used GitHub Issues to track tasks, assign responsibilities and monitor each other's progress.

A work log was maintained to document the team's activities over the course of the project. We've documented our meeting discussions through Minutes of Meeting (MoM) records. Links to code files, supporting evidence of progress are included in Appendix (sec. **B**).

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1 Abstract

*“Without data, you are just another person with an opinion.”
– W. Edwards Deming*

E-commerce platforms generate large volumes of transactional and operational data that can provide valuable insights into customer experience, logistics performance and seller outcomes. This project analyzes data on an online marketplace in Brazil that connects sellers and customers. The [dataset](#) covers orders, deliveries, payments, customer reviews, and seller onboarding records over a two-year period. The analysis examines ways to make the marketplace grow by conducting a vertical study of sellers, product categories, and regional demands in relation to customer satisfaction and suggests interventions for the company to tackle the issues in each head. While the company had an overall good track record of deliveries, relatively poor ratings for certain sellers and products are correlated with delayed deliveries. Such sellers and products that needed intervention were identified to provide individual attention by tracking them through dashboards. There is also a lopsided distribution of sellers, with nearly 60% of them in one state, while some states with a sizable number of customers do not have sellers for high-demand products. Such states and corresponding products were identified to find sellers through appropriate market funneling channels. To relieve sellers facing supply pressure, leading to delayed delivery, the study identified alternative sellers for them using geolocation data. Overall, the study provides actionable suggestions to improve operational efficiency and bridge the gaps between the customer-seller ratio of the states.

2 Introduction

2.1 Project Background

An anonymized marketplace platform has started spreading its wings in Brazil in the summer of 2016-2017. By a year, it reached almost all states of Brazil and expanded its list of sellers and product categories. However, this growth, mostly due to the exuberance of youthfulness, must be sustained. We were given records of orders placed in the anonymized company between 2016 and 2018, along with information about sellers, products, customer reviews, and a few other heads, to analyze and suggest ways for the company to grow without adversely affecting customer satisfaction.

2.2 Business Problem

As a regulator of a marketplace, with profit as a motive, the company must keep its two pillars – sellers and buyers – happy. To grow, the company must find trustworthy sellers and bring their products to buyers on time. In this transaction, the regulator must satisfy sellers by giving them a good return while helping buyers find products of good quality at an affordable price. This balance can be spoiled by late deliveries due to logistics gaps or/and by unfulfilled promises of sellers made to customers. Either of these will drive away customers, making the marketplace redundant. So, the business problem that the regulator, especially a young one, faces is preventing the marketplace from breaking down due to the above-mentioned reasons.

2.3 Project Objective

The primary aim of this analysis is to identify the ways for the company to sustain its growth while entrenching its good practices and weeding out bad practices **without compromising customer satisfaction**. Rather than trying to find a silver bullet in the dataset for the company’s growth, this analysis tries to find small incremental improvements across various heads so the company achieves sustainable growth and thrives for a long time. The project examines the complete e-commerce journey, including seller performance, order handling, delivery outcomes, regional demand and customer reviews, to identify actionable measures to achieve this objective.

2.4 Main Research Questions

- How do order volume, status, and customer reviews change over time and space?

- How strongly are delivery delays associated with poor reviews?
- Which product categories top in volume, revenue and manage risk well and not so well? Identify product categories that need intervention.
- Which sellers need intervention, and who provide useful benchmarks?
- Which states offer attractive demand while requiring delivery improvements?

3 Data Sources and Dataset Description

3.1 Dataset Overview

The dataset is made up of 11 related tables covering orders, order items, payments, reviews, products, sellers, customers, geolocation, and the seller sales funnel (marketing qualified leads and closed deals). Each table represents a different part of the marketplace process, from a seller being recruited onto the platform to a customer leaving a review after delivery.

3.2 Source Tables

- **orders** – one row per order, with status and timestamps for purchase, approval, carrier handoff, and delivery.
- **order_items** – one row per item within an order, including price, freight cost, and the seller and product involved.
- **payments** – one row per payment transaction for an order, including payment type and number of installments.
- **reviews** – one row per customer review, including score, optional comment text, and review dates.
- **customers** – customer location information linked to each order.
- **sellers** – seller location information.
- **products** – product category and physical dimensions.
- **product_category_name_translation** – maps Portuguese category names to English.
- **geolocation** – latitude and longitude by zip code prefix.
- **marketing_qualified_leads (mql)** – leads that entered the seller sales funnel, with origin and first contact date.
- **closed_deals** – leads that converted into active sellers, with onboarding details declared at sign-up.

3.3 Table Relationships and Keys

`order_id` links orders, order_items, payments, and reviews. `customer_id` links orders to the customers table, which in turn links to `customer_unique_id` for identifying repeat customers across multiple orders. `seller_id` links order_items to the sellers table and, where applicable, to closed_deals. `product_id` links order_items to products, and `product_category_name` links products to the category translation table. `mql_id` links marketing_qualified_leads to closed_deals for sellers that converted. Zip code prefixes link customers and sellers to approximate coordinates in the geolocation table, though this mapping is many-to-one and requires aggregation before use.

3.4 Data Grain and Unit of Analysis

The unit of analysis varies by table. The orders table is at the order level. The `order_items` table is at the order-item level and one order can contain multiple items from different sellers. The payments table is at the payment-transaction level and similarly one order can be paid in multiple installments or split across payment methods. The reviews table is at the review level, with one review generally corresponding to one order. The `closed_deals` and `marketing_qualified_leads` tables are at the seller-lead level. Aggregation to the

order, seller, or state level is performed where needed for specific parts of the analysis, and each result specifies which grain it applies to.

3.5 Variable Description

- `order_status` – current status of the order (e.g., delivered, canceled, shipped).
- `order_purchase_timestamp` – when the order was placed.
- `order_approved_at` – when payment was approved.
- `order_delivered_carrier_date` – when the order was handed to the logistics carrier.
- `order_delivered_customer_date` – when the order was delivered to the customer.
- `order_estimated_delivery_date` – delivery date estimate shown to the customer at purchase.
- `price` – item price, excluding freight.
- `freight_value` – shipping cost for the item.
- `payment_installments` – number of installments used to pay for the order.
- `review_score` – customer rating from 1 to 5.
- `review_comment_title` / `review_comment_message` – optional free-text review content.
- `lead_type` – category describing how the lead was classified during the sales process.
- `lead_behaviour_profile` – classification of the lead's behavior during the sales process.
- `has_company` – whether the seller had a registered company at sign-up.
- `has_gtin` – whether the seller's products had valid barcodes (GTIN) at sign-up.
- `declared_product_catalog_size` – catalog size the seller reported at sign-up.
- `declared_monthly_revenue` – monthly revenue the seller reported at sign-up.
- `origin` – marketing channel through which the lead was acquired.

4 Data Integration and Preparation

4.1 Data Loading and Data Standardization

The e-commerce and marketing funnel datasets were loaded and preprocessed using Google Colab and the data was standardized across all tables to make it easier to join and analyze.

- Non-ID text and categorical values were converted to lowercase.
- Underscores were replaced with spaces.
- Newline and carriage-return characters were removed.
- Date and numeric fields were converted to appropriate formats.
- ID columns were kept unchanged.

4.2 Categorical and Text Cleaning

The text field and categorical field were standardised before the datasets were joined. This was crucial to avoid formatting differences that can cause the same value to be treated as different values later during the joining phase, filtering, grouping, and aggregation. All non-ID text and categorical values were converted to lowercase. Underscores were replaced with spaces, and the newline characters and (`\r` and `\n`) were removed. These changes

helped us maintain a consistent format across all the categorical fields in the dataset. Identifier columns such as `order_id`, `customer_id`, `product_id`, and `seller_id` were not modified because their original values were necessary for matching records between datasets. The cleaning was focused only on the format of the values and did not change their underlying meaning. So, this provided a consistent set of categorical and text fields for the subsequent data integration and analysis.

4.3 City Name Standardization

City names needed additional cleaning because the same location appeared in different forms across the customer and geolocation data tables. The differences included inconsistent spacing, accented characters, and certain variations in how city names were written. The city name values were manually inspected to identify these inconsistencies and determine whether different forms referred to the same location. Based on this review, city names were standardized by removing accents, correcting spacing, and cleaning encoding-related characters. For example, `saopaulo`, `sãopaulo`, and `safo paulo` were standardized to `sao paulo`. Similarly, `4o. centenario` was standardised to `4o centenario`. This step was particularly important for matching customer records with the geolocation dataset. Without standardization, different representations of the same city could have been treated as separate values, reducing the accuracy of location-based matching.

4.4 Missing-Value Handling

Missing values were checked all across the datasets. Empty strings and other null-like values were treated as missing values (`NA`) so that they could be handled consistently later. Existing missing values were not imputed since there was no reliable information available to replace them. For general fields, missing values were represented as `NA`, while missing values in date and timestamp fields were represented as `NaT`. No general imputation method was used, as assigning values without a clear basis could introduce information that was not present in the original data originally. For fields like dates and timestamps, keeping missing values as `NaT` was important because filling missing dates with estimated values could introduce inconsistencies into these calculations. Overall, the approach taken was to preserve missing information rather than artificially replace it.

4.5 Date and Timestamp Standardization

All date and timestamp columns were converted to a standard format so that it stays consistent before the datasets were joined and analysed. This process allowed the dates to be compared correctly. The order timestamps were then checked for logical inconsistencies between the purchase, approval, carrier, delivery, and estimated delivery dates. The main issue identified was that, for 166 orders, the carrier date occurred before the purchase date. These 166 cases were investigated further rather than being removed immediately. 160 cases were within two hours, while 6 cases were more than two hours early, including 2 cases that were more than 24 hours early. For the invalid carrier dates, imputation was not implemented because replacing them with estimated dates could introduce artificial information and distort the results of time series based analysis. Instead of that, the invalid carrier dates were changed to `NaT`, while the affected orders were retained in the dataset. A separate flag was also created to identify records where the carrier date had been found to be invalid. This approach allowed the original order records to be preserved while preventing unreliable timestamps from affecting calculations.

4.6 Geolocation Aggregation

The geolocation dataset had multiple latitude and longitude values for the same ZIP code prefix. Since the ZIP code prefix column should be used to link geolocation information with customer and seller tables, joining the uncleaned geolocation data directly could create multiple matches for a single customer or seller and increase the number of records. To prevent this problem, the geolocation data was aggregated at the ZIP code prefix level, so that each ZIP code prefix had only one representative location. The median latitude and median longitude were used as the representative coordinates. This produced a geolocation table with one record per ZIP code prefix, which could then be safely joined to the customer and seller datasets without unnecessarily duplicating records.

4.7 Product Category Translation

The product dataset contained category names in Portuguese, and the category translation table provided their English equivalents. The two datasets were joined using `product_category_name`. Initially, 13 product-category rows did not have a matching English translation, representing 0.04% of the product-category join rows. These rows belonged to two unique categories: `pc gamer` and `portateis cozinha e preparadores de alimentos`. Rather than leaving these categories as missing values, the two missing entries were added to the translation table using verified English labels. This method ensured that all affected product records had a corresponding English category in the final dataset.

4.8 Data Joining and Master Table Construction

The cleaned datasets were combined using their Foreign Keys, such as `order_id`, `customer_id`, `product_id`, and `seller_id`. LEFT JOINS were used so that the main order records were preserved even when their related information was missing. The main marketplace tables were joined through orders, order items, customers, products, sellers, payments, and reviews. The marketing data was also linked through `mql_id` and `seller_id` to connect marketing leads with sellers and their marketplace activity. After joining the datasets, the final master table had 112,650 rows and 67 columns. The table was maintained at the order-item level, using the combination of `order_id` and `order_item_id` as the Primary key. This structure provided a single dataset containing the customer, product, seller, order, payment, review, location, and marketing information required for the analysis.

5 Data Quality Assessment

5.1 Duplicate-Key Checks

The primary key for the master table was identified as the combination of `order_id + order_item_id`, as this composite key uniquely identifies each order item record in the master table. The Composite Key was validated before performing the joins to make sure that the joining process would not introduce duplicate or incomplete records. The validation showed that there were 0 duplicate `order_id + order_item_id` combinations and 0 missing values in the key fields. By Performing Duplicate key checks it had confirmed that the composite key uniquely identifies every record in the master table and that the master table maintains key-level integrity after the joining

5.2 Join Integrity Checks

Join integrity checks were performed to verify that the relationships between the datasets were correctly established during the construction of the master table. The main marketplace and marketing joins had 0 unmatched records except for the geolocation joins had 278 unmatched customer records and 7 unmatched seller records. These records were retained, with the unavailable location fields left as missing. The product category translation join initially had 13 unmatched rows across two categories. These were manually reviewed and added to the translation table using verified English category names, so no category information remained missing.

5.3 Missing Values and Unmatched Records

Missing values were checked across all columns and Primary key fields. No missing values were found in the validated composite key, and the naturally existing missing values had already been checked for. Some missing values resulted from unmatched join records. For geolocation, the cleaned table contained 19,015 unique ZIP-code prefixes, but 278 customer records and 7 seller records had no matching ZIP-code prefix. These records were retained, with only the location fields left missing (NA). Unmatched values only affected geographic information and did not affect the main order and sales records, so artificial values were not added.

5.4 Final Dataset Validation

After all preprocessing and joins, the final master table was validated for row count, column count, key uniqueness, and data grain. The final dataset contained 112,650 rows and 67 columns which matched the original order-item row count. No duplicate `order_id`, `order_item_id` keys were found, confirming that the joining process did not introduce duplicate records. The final grain was preserved as 1 row = 1 order item. This validation confirmed that the Joined dataset was structurally consistent and ready for further analysis.

6 Methodology

Keeping in the mind the vastness of data, the team decided to carry board studies under various categories to understand the marketplace. This decision shaped further analysis, and steps followed were listed below:

- First, more than 20 questions were listed and categorized under orders, sellers, products and ratings. Each member of the team volunteered to carry out exploratory data analysis under each head. Besides finding regular descriptive statistics for each head, the team member answered the questions listed in that particular head.
- The team then discussed results of EDAs under each category. Broad issues, including relationship between delayed delivery and review rating and lopsided customer-seller ratios in some states, were identified.
- Each member then extended analysis by carrying out cross-head study. For example, product categories that were consistently being reviewed low were identified.
- Then three areas where the study could contribute were identified and appropriate solutions to each problem were reviewed. Among them, the actionable ones were suggested as measures.

6.1 Analysis Framework

EDAs carried out by each member are saved as different notebooks in GitHub repo (check appendix C). **Rationale behind various decisions** are listed below:

‘Late’ definition: Orders were joined with reviews and customers, then any order missing a delivery date, an estimated delivery date, or a purchase timestamp was dropped as keeping them would have quietly dragged the averages around. This left only delivered, reviewed orders – the set where “was it late” and “was the customer happy” can actually be compared. Then, two simple flags were built: `delay_days` (delivered date minus estimated date) and `is_late` (`delay_days > 0`). All the charts and comparisons use these same two columns. This keeps the analysis consistent throughout the notebook and avoids defining “late” in different ways.

Bucketing delivery days: For delivery delay vs. review score, rather than jumping straight to a correlation number, `delivery_days` was binned into buckets (0–3, 4–7, 8–14, etc.) and average review score was plotted per bucket. Bucketing makes the trend visible in a way a single correlation coefficient wouldn’t, and it also exposes where the drop happens rather than just confirming that one exists.

Delivery comparison based on time: A late vs. on-time comparison of both average score and the share of 1–2 star reviews were shown since a small drop in average score can hide a much bigger jump in genuinely angry customers. A t-test on late vs. on-time review scores was also done.

Delivery comparison based on space(state): The same-state vs. cross-state shipments was also checked. If same-state orders are both faster and better reviewed, that’s a cheap, non-obvious lever for the business (favor local sellers/warehousing) rather than just “ship faster” which isn’t always actionable on its own.

Regional demand and delivery risk: For this, customer states were ranked by revenue and volume first, since “which states matter” has to come before “which states are risky”. A minimum order-count threshold (30 orders for state-level stats, 20 for seller→customer routes) was added before ranking anything by late rate. The main goal is to find states that have both high sales and high delivery risk. These states are the most important to fix because improving them could have the biggest impact. Seller-to-customer routes were also looked at to check whether the problem is spread across a whole state or comes from a few specific shipping routes. This helps decide whether to improve shipping across a state or fix individual routes.

Aggregation of seller data: The seller table only identifies sellers through `seller_id`, so a seller-level summary was first created from `order_items`. The data was grouped by `seller_id` to calculate revenue, order count, item count, and freight value. This converted the line-item data into one row per seller for further analysis.

Order count threshold for sellers: A minimum order-count threshold was applied before comparing late rates. Sellers with very few orders can have unstable rates; for example, one late order out of two results in a 50% late rate. The distribution of delivered orders per seller showed a rapid decline in seller counts as order volume increased. Based on this distribution, sellers were required to have at least 20 delivered orders to be included in late-rate comparisons. The threshold was applied to review analysis as well.

Criteria for seller intervention candidates: In addition to the order-count threshold, the 75th percentile of each metric was used to define “high” revenue, late rate and low-review rate. Sellers exceeding all three thresholds – high revenue, high late rate and high low-review rate – were classified as the intervention watchlist. Sellers with high revenue but late and low-review rates below the corresponding thresholds were used as the healthy benchmark group.

Criteria for product intervention candidates: High risk categories = (high revenue OR high volume) AND weak reviews. High revenue and volume mean greater than median value. Similarly, weak reviews mean average review score is less 4 or at least 10% of orders under the category was given poor reviews.

Criteria for seller suggestion to improve customer-seller ratio in certain states: A weighted score was devised with four factors: Customer demand for product category (should at least 50 over the period); Local seller availability (either 0 or 1); Late delivery rate for the product category is at least 25% and review ratings have at least 20% of poor reviews.

6.2 Metric Definitions

Customer satisfaction is measured by the post-purchase review score (1–5), with a low review defined as a score ≤ 2 and a high review as 4–5. **The delivered rate** is the share of orders with a delivered status and a recorded customer-delivery date. **The late rate** is late orders divided by delivered orders. **Revenue and freight** are the sums of item-level price and freight value. **The acquisition win rate** is closed deals divided by marketing-qualified leads.

6.3 Aggregation Levels

Metrics are computed at the grain that matches the underlying field. Order-level metrics (delivery time, lateness, review score) use one row per order. Item-level metrics (revenue, freight, category and seller contribution) use one row per order item because price and freight are recorded per item. Multi-item orders are counted once so they are not double-counted in order-level KPIs. These base grains are then rolled up to weekly, category, seller and state levels for the dashboards.

6.4 Derived Variables

Several fields were engineered from the raw timestamps and joins: `delivery_days` (customer delivery – purchase), `delay_days` (customer delivery – estimated date), and the flags `is_delivered`, `is_late` (delivered with `delay_days` > 0), `low_review` (score ≤ 2) and `one_star` (score ≤ 1). We also derived `same_state` (customer state = seller state), `freight_ratio` (freight $\div$ price), and calendar fields - `purchase_week` (the Monday of the order’s week), `month`, `weekday` and `year` – to support the time-based and geographic views.

7 Data Analysis and Key Findings

7.1 Order Journey & Satisfaction

Most orders in the marketplace were successfully delivered, but non-delivered orders are still important for understanding why some placed orders were not completed. Monthly and weekly patterns give us a baseline to compare with later segments and see if they follow the seasonal trends.¹

¹Note: Plots had to be fitted to the available space, may require zooming for clarity. The complete set of plots for each subsec. is available in the respective notebooks in our repository (check C).

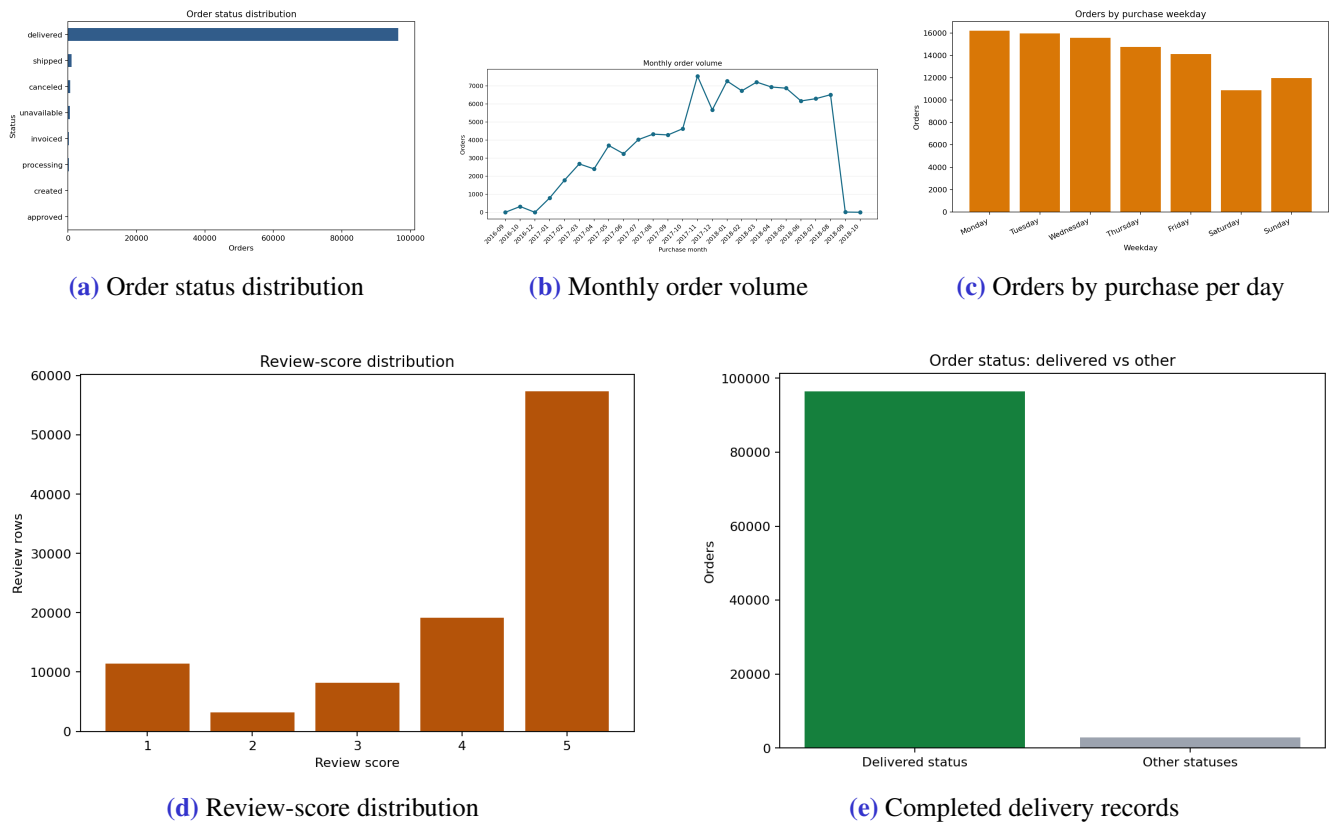


Figure 1: A few plots from the order journey and satisfaction analysis

- The highest number of orders was recorded in November 2017, with 7.5k orders.
- The weekday pattern peaks on Monday. It had the most orders overall, with $\approx 16.2k$ orders.
- Five-star reviews make up the largest group but the high share of low ratings still needs further investigation.

7.2 Delivery Delay Impact

Even though only around 10% of orders were delivered late, the absolute number is greater than 10,000. Besides finding the product categories and sellers that report most delays, it is also essential to understand if the delays were impacting customer reviews.

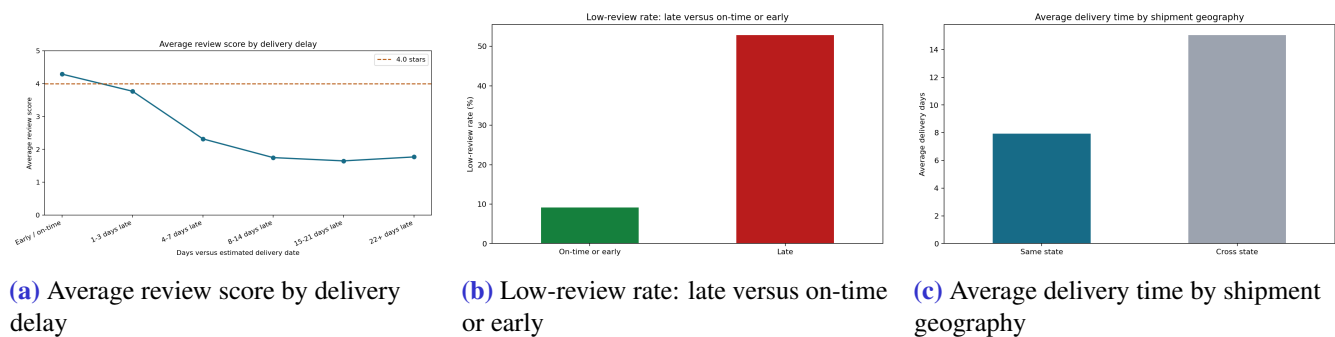


Figure 2: A few plots from the delivery delay analysis

- Delivery reliability is the strongest operational signal for satisfaction. Average review score drops below 4.0 stars once delivery exceeds 21 days from purchase.
- Late deliveries show a low-review rate of 62.4% vs 9.3% for on-time – a 6.7x increase.
- Same-state shipments deliver in 7.5 days on avg vs 14.6 days cross-state.

7.3 Marketing Funnel

The marketing funnel was analyzed to see how 8,000 MQLs were converted into closed deals and how many of the acquired sellers could be linked to marketplace activity. Of the 8,000 MQLs, 842 had a matching closed-deal record, giving us an observed conversion rate of 10.53%.

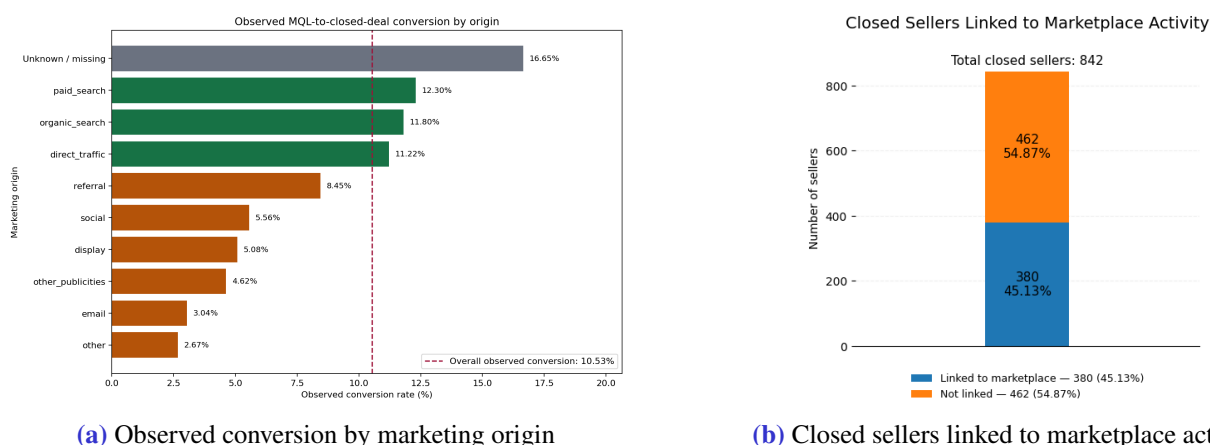


Figure 3: A few plots from the marketing funnel analysis

- 380 of 842 closed sellers (45.13%) could be linked to marketplace records.
- Paid search and organic search converted above the overall 10.53% benchmark, with organic search generating the highest number of closed sellers (271).
- Social and email showed low observed conversion rates of 5.56% and 3.04%, with 1,275 and 478 MQLs respectively.

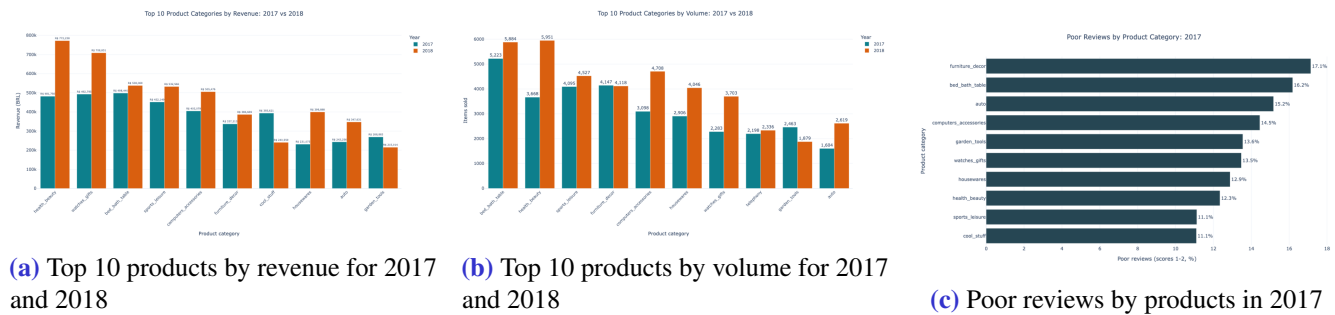
The conversion rates were different across marketing origins. The 842 closed sellers were then checked against the marketplace data using `seller_id`. This linked 380 sellers (45.13%) to marketplace activity, while 462 had no matching marketplace record. For the 380 linked sellers, we checked sales, sales value, number of customers, seller activity, and customer reviews.

These measures were also compared across marketing origins. Since only 380 sellers could be linked, the marketplace analysis covers only part of the closed sellers. The results therefore show differences and associations, but do not prove that a marketing origin caused better seller outcomes. The funnel is mainly used as supporting evidence, with social and email as areas for further investigation and the Unknown / missing group showing an attribution issue.

7.4 Product Category Analysis

In the analysis of products, top categories in terms of revenue and volume were identified. Then review ratings for these categories were found before identifying the intervention candidates by the above-mentioned criteria. The

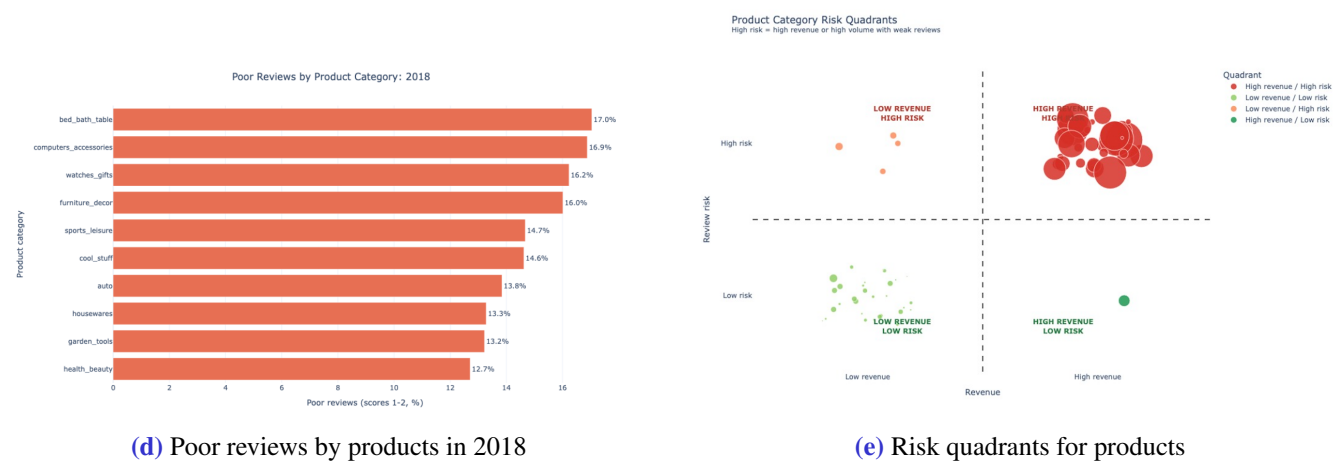
goal was to see which categories are actually growing, which ones are large but risky, and which ones deserve more attention or investment going forward.



(a) Top 10 products by revenue for 2017 and 2018

(b) Top 10 products by volume for 2017 and 2018

(c) Poor reviews by products in 2017



(d) Poor reviews by products in 2018

(e) Risk quadrants for products

Figure 4: A few plots from the product category analysis

- Health beauty category led in terms of revenue and volume in 2018 while Bed bath table led in 2017
- Revenue and volume of top-10 product lists overlap by 90%, but are not identical
- Telephony appears only in the combined volume list of top 10 while cool stuff appears only in the combined revenue list of top 10
- Housewares had the strongest January-August revenue growth of 73.1% while Auto had the strongest January-August volume growth of 63.3%
- High sales do not guarantee better reviews. Bed bath table has poor-review rate of 17.0% while watches gifts category has high revenue but 16.2% poor reviews

7.5 Seller Performance Analysis

Analysis of volume, revenue of orders among sellers, their delivery performance and customer reviews were combined to identify sellers who required intervention. Sellers were grouped by revenue, order count, late-delivery rate and review scores to separate high-value sellers who are performing well from those who are not. Seller state was also checked to see if geography plays any role in delivery performance.



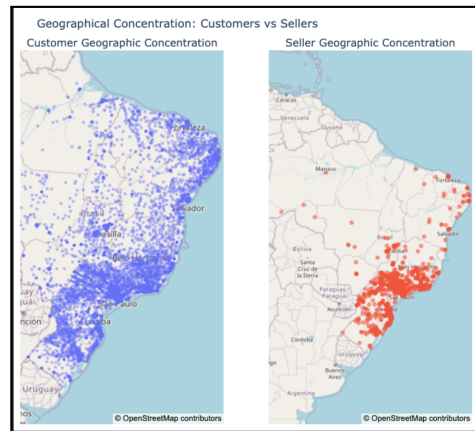
Figure 5: A few plots from the seller category analysis

- Fig. 5c shows that several high-revenue sellers have lower average reviews than the overall sellers. They are commercially important & their customer experience need more attention.
- High demand is NOT always high experience! Some high-revenue sellers have low review scores (< 3.8), including sellers with scores around 2-3. They also need further monitoring.
- In chart 5b, order volumes for sellers are highly uneven. The median seller generated six unique orders and R\$821.48 in product revenue, while the highest seller generated 1,854 orders and R\$229,472.63 in revenue
- The highest-revenue sellers contribute substantially different numbers of orders. For eg., the second-highest revenue seller generated approximately R\$222.8K from 358 orders, while another R\$200.5K from 1,806 orders.
- Seller contribution varies substantially by both order volume and revenue. High-volume sellers do not necessarily generate the highest revenue
- Seller delivery performance varies by geography, but state-level comparisons must be interpreted cautiously as seller counts differ substantially.

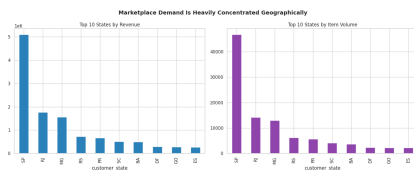
7.6 Regional Demand and Risk

The preliminary regional analysis showed lopsided distribution of sellers and customers. Closer to 60% of sellers and 40% of customers are from Sao Paulo. However, certain states showed healthy demand for high revenue products and either did not have sellers or had only one seller. Such states and products were identified and shown in Fig 6(e). The move will also help improve customer-seller ratio. The company should find sellers for these products in

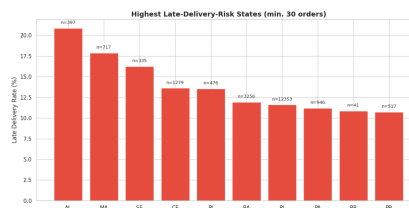
respective states on priority basis as same-state sellers tend to show better delivery record.



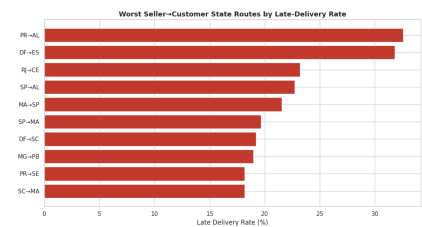
(a) Geographical concentration of customers and sellers



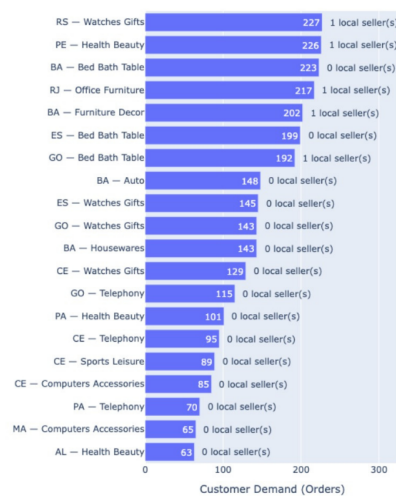
(b) Top 10 states by Revenue and Volume



(c) States with Highest Late-Delivery Risk



(d) Interstate Routes with High Late Delivery Rate



(e) States and products with no or one seller

Figure 6: A few plots from the demand risk analysis

- Customers demand and sellers are heavily concentrated in São Paulo
- Demand Concentration: Top 3 states drive 63.4% of revenue – SP, RJ, MG
- Risk concentration: Highest late-delivery-rate states (reliable sample with min. 30 orders): AL, MA, SE

- Priority Zone (High demand AND High risk): CE, BA, RJ, PA, ES, PE
- Worst Routes: PR → AL (shows a 32.6% late rate on 43 orders.)

8 Business Implications and Recommendations

- The insights from sec. 7.1,7.2 show that delivery performance has a direct impact on customer satisfaction, making delivery reliability an important area for the business to address. The company can use a delivery guardrail of around 20 days to identify orders at higher risk of receiving poor reviews and trigger proactive customer communication.
- Since cross-state deliveries are slower and more likely to create review risk, logistics improvements should focus on these routes, with priority given to high-demand and high-risk states (sec. 7.6 fig. 6d).
- The business should actively search for sellers for products in good demands in selected states where there are no active sellers or only a few of them (sec. 7.6 fig. 6e). Regional warehouses, carrier improvements and same-state seller incentives can be considered for the priority zones and worst-performing routes.
- To relieve sellers who are struggling to deliver products on time, the business can use details of alternative sellers given the dashboard (link given in the appendix) to divert the orders. The sellers were identified using geolocation data so that they are closeby.
- The business should also track delivery delay against the promised estimate, as this better reflects the customer's experience than delivery time alone. For this purpose, the company can concentrate on priority sellers and products identified for intervention in (sec. 7.4 fig. 4e) and (sec. 7.5 fig. 5c) respectively.
- The company should better maintain sellers acquisition data and work towards converting more MQLs into sellers (sec. 7.3 fig. 3a). It also needs to bring converted leads into marketplace at the earliest. Rather than blindly adding sellers, the business should select priority routes and states to address customer-seller ratio discrepancies in some states.
- For low-volume or/and low-revenue states, the business should continue monitoring performance rather than making decisions based on limited data.

9 Conclusion

The study analyzed problems of the business under three heads: geography, products and sellers. Review ratings were used as the underlying factor or a common theme that bind together the vertical analyzes and key criteria deciding on the recommendations made to the business. The study identified sellers and products that need interventions. It found the problematic routes that needed further study and states that needed sellers of particular products. To ease pressure on sellers who struggle to meet the demand, alternative seller in the area were identified using geolocation data and made available in the dashboard. These result-orientated suggestions, if implemented, would help the business achieve its next level of growth and improve customer satisfaction, paving the way for long-term growth.

10 Limitations and Future work

- During the marketing funnel analysis, the closed sellers were matched to sellers in the e-commerce marketplace dataset using `seller_id`. Only 380 of the 842 closed sellers (45.13%) had a matching seller record with marketplace activity. The remaining 462 sellers had no matching marketplace records, so their activity in the e-commerce dataset could not be analyzed. Because of this limitation, the analysis could only show an association between marketing origin and marketplace seller performance, but not causation or best marketing method. The **Unknown / missing** marketing origin should be reduced through better tracking.

- The geolocation data was aggregated to ZIP-code prefixes so that it could be joined with customer and seller records. This reduced the geographic detail available for the analysis. In addition, 278 customer ZIP-code prefixes and 7 seller ZIP-code prefixes could not be matched to the dataset, leading to limited analysis.
- The marketplace analysis is also based on historical data ending in 2018, so the findings may not represent current marketplace performance.
- There is also no information about warehouses, movement of freight per se and transporters, limiting the analysis from finding the exact reasons delay in deliveries.
- There is no general information about the business and explanation of variables in dataset, forcing analyzers to make multiple assumptions and bringing down the quality of work.
- The criteria defined for finding various threshold and making classifications were statistical and analytical in nature rather than requirements of the business.
- With the improved dataset, the future study can concentrate on finding solutions to micro-problems affecting delivery.
- Similarly, improved MQL dataset would help streamline the seller acquisition process and will help the business grow in future.

Acknowledgments

We are thankful to Prof. Venkatesh Rajmanickam for the lessons and Rohit Sir for his support and guidance. The report was compiled on Overleaf, using a $\text{\LaTeX}$ template adapted from the [UPIIAP-IPN report](#).

Appendix

A Interactive Marketplace Dashboard

Alongside the static analysis, we built a live dashboard so the marketplace's performance. It's a web application built with Next.js and Tailwind CSS, with all of the charts drawn using Apache ECharts. It is deployed and accessible at <https://pubmed-dui-necessity-barnes.trycloudflare.com>.² The dashboard has two main controls available across all pages: a weekly slider and a state filter. The weekly slider allows users to select a specific week while the state filter can be applied using a drop-down or by selecting a state on the map. The KPIs, charts and tables update based on the selected week and state. Charts also support zoom and full-screen viewing, and the dashboard includes a light/dark mode option.

To keep the story simple, the dashboard has three main sections:

- **Overview:** Shows weekly KPIs including total orders, delivered orders, late deliveries, average review score, week-on-week change and the best week. It also includes cumulative order volume, state-wise order volume, a state-level late-delivery-rate map and the top and bottom three categories and sellers for the selected week.
- **Product:** Shows total and delayed orders by category, weekly changes in average review scores and a risk quadrant comparing category revenue and customer reviews. A table highlights categories that require attention.
- **Seller:** Shows sellers with high order volume but weaker delivery or review performance. It includes seller-level order and delay information, review trends, and a comparison of seller revenue and customer satisfaction. For sellers requiring attention, the dashboard also identifies a nearby healthy seller offering the same category as a possible alternative.

Backend: The dashboard uses a PostgreSQL database connected to the application through API routes. The selected week and state are passed to the API, which runs parameterized SQL queries and returns the required data for each chart.

²we plan to keep the deployment active till the result release.

The database uses the cleaned project dataset. The PostgreSQL queries 10 objects: 2 base tables – `order_base` and `item_base`, built from the raw tables by the cleaning pipeline and 8 derived views/tables – `weekly_overview`, `weekly_category`, `weekly_seller`, `category_stats`, `seller_stats`, `intervention_sellers`, `tracked_categories` and `seller_alternatives`. Materialized views are used for frequently required calculations, including weekly summaries, category and seller statistics, intervention lists, healthy-seller lists and nearest alternative sellers. The nearest alternative seller information is calculated using the Haversine distance between seller locations. Materialized views were used to reduce repeated calculations and improve dashboard response time. Database connection pooling and query timeouts are also used to manage database requests.

B Three-Week Progress Log

The project work was completed through regular team discussions and weekly progress. Each week, specific goals and targets were set to ensure timely completion of the assigned tasks. At the beginning, we discussed the project requirements, identified multiple questions that could be answered using the dataset. All individual datasets were merged into one sheet. We went through the data and started the initial exploratory analysis.

During the first stage, the team reviewed the dataset, identified missing values and data quality issues and discussed the possible analysis questions. Everyone came up with their own EDA and questions and these were then compared among team members to identify overlaps and decide the final areas of analysis.

The EDA work was divided into 12 GitHub issues, with individual responsibilities assigned to team members. Each member completed their assigned analysis and the results were discussed and compiled as a team. In the final stage, the team reviewed the completed analysis and focused on the dashboard, technical report and presentation. Feedback was discussed over meets and changes were made to improve the dashboard layout, visualizations and clarity of insights.

Regular meetings were held throughout the project to track progress, discuss issues and decide the next steps. **The [meeting minutes](#), calendar records, screenshots of internal discussions, collaborative work on Overleaf, our [internal GDoc](#) screenshots showing development of questions and other evidence of project progress** such as closed issues, PRs can be viewed at <https://drive.google.com/drive/folders/1f9ooSDGnHP8EJ019VLVYTvDjbMpLFT7?usp=sharing>.

C Reproducibility

All code and files are available at <https://github.com/22f3002680/Dataviz-project-Group3>.

Key branches:

- **main** – contains the full code-base for the dashboard deployed via Cloudflare.
- **project-materials** – contains the notebooks and working files used during analysis and development (please check [readme](#)).

(We had initially published a static dashboard via GPages to get a basic overview which is also available in the gh-pages branch.) Also, our presentation can be accessed [here](#).

Step-by-step instructions to set up and run the full pipeline:

```
# First clone the repo
git clone https://github.com/22f3002680/Dataviz-project-Group3.git
cd Dataviz-project-Group3

# Install the Python dependencies
pip install -r requirements.txt
```

```
# Build the data and start the database (PostgreSQL in Docker)
python3 scripts/build_metabase_tables.py
python3 scripts/build_seller_geo.py
bash metabase/setup_stack.sh

for f in marts geo_marts materialize; do
  docker cp metabase/$f.sql dvd-postgres:/$f.sql
  docker exec dvd-postgres psql -U olist -d olist -f /$f.sql
done

# Run the web app
cd webapp
cp .env.example .env.local
npm install
npm run dev

Open http://localhost:3001
```

■ ■ ■